

Course Title	Database Systems Lab				
Course Code	CC-215-L				
Credit Hours	1 (0,3)				
Category	Computing Core				
Prerequisite	None				
Co-Requisite	None				
Follow-up	DI-324: Database Administration and Management				
Course Introduction	The course aims to introduce the Structured Query Language (SQL). It covers the set of commands related to Data Retrieval, Data Definition Language (DDL), Data Manipulation Language (DML), Data Control Language (DCL). It will be followed up by procedural flavor of SQL (PL/SQL).				
Course Learning Outcomes (CLOs)	At the end of the lab, the students will be able to:	BT	PLO		
	CLO1: Basic understanding of different commands of SQL, and the fundamental knowledge of PL/SQL	C2 (Understand)	1,2		
	CLO2: Writing queries as per the given requirement	C3 (Apply)	3,4,5		
	CLO3: To evaluate query processing	C5 (Evaluate)	3,4,5,7		
Syllabus	Introduction to SQL environment: Writing Basic SQL Statements; SELECT Statement: Arithmetic Expressions, Operator Precedence, Null Value, Column Alias, Concatenation Operator, Comparison operators, Logical operators, ORDER BY clause Single row functions: character functions, number functions, date functions, type conversion functions. Multi row Functions: Sum, Average, Standard deviation, Variance, Subgrouping of data, Group by clause, Having clause, CUBE, ROLLUP, Analytic functions. Join: Cross product, natural join, Equi-join, Non equi-join, left outer-join, right outer-join, self-join. Subquery: use of subquery, subquery syntax, Multiple column subqueries, pairwise comparison, Non-pair wise comparison, Null Value in a subquery, Subquery in From Clause. Creating and Altering database tables: Create table statement, Defining table and column level constraints (NOT NULL, PRIMARY Key, FOREIGN Key, UNIQUE Key, CHECK), Alter table statement, drop statement, Data Manipulation: Insert, Update, Delete statements. SQL Objects: Views, Sequences, Indexes. User Management: Create user, user privileges, user groups, Grant, Revoke statements. Introduction to Procedural SQL (PL/SQL): Sections of a PL/SQL block, Variable declaration and initialization, SELECT statement in PL/SQL, Arithmetic expressions, Selection, Repetition, Exception Handling, Cursors, Stored Procedures and Functions, Database Triggers				
Suggested Instructional/ Reading Material	1. Introduction to Oracle 9i: SQL 2. Michael McLaughlin, Oracle Database 11g PL/SQL Programming, 1 st Edition, McGraw-Hill Education, 2008, ISBN: 0071494456. 3. Jason Price, Oracle Database 11g SQL, McGraw Hill. ISBN: 0071498508.				
Assessment Plan	Criteria	Marks	CLO1	CLO2	CLO3
	Labs	70	30	30	10
	Quizzes	15	10	05	00
	Lab Project	15	05	10	00
	Total	100	45	45	10

Lab#	Topics
1	Installation of Oracle / MS SQL Server, Introduction to SQL Commands
2	Data Retrieval, SELECT statement, Arithmetic Operators, Relational Operators, Logical Operators, WHERE clause
3	Special operators (BETWEEN, IN, LIKE, IS NULL), Use of DISTINCT, Column Alias, ORDER BY clause
4	Quiz#1 , SQL Functions, Single row functions, Handling NULL values (NVL), Character functions
5	Functions for numeric and date data types, Type conversion functions
6	Decode and CASE expressions, Multi-row (group) functions, Group by clause, Having clause
7	Quiz#2 , Revision tasks, Use of ROLLUP, CUBE, Introduction to SQL Joins
8	SQL Join Types: Cartesian Join, Inner or Equi Join, Outer Join (Left, Right, Full), Non-Equi Join, Self-Join
9	Subqueries, Single-row and Multiple-row subqueries, Use of IN, ANY, ALL operators in subqueries, Multiple-Column Subqueries, Correlated Subquery,
10	Quiz#3 , DDL commands, Creating database tables, Applying table and column level constraints (PRIMARY KEY, NOT NULL, UNIQUE, FOREIGN KEY, CHECK, DEFAULT), ALTER command, Data Dictionary views for tables and constraints management, Disabling and Enabling of constraints, Renaming a table or column, Drop, rename, and truncate tables,
11	Creating a table using subqueries, DML commands (INSERT, UPDATE, DELETE), Virtual Tables (Views), Indexes, Sequence, Synonyms
12	Quiz#4 , Revision, Controlling User Rights & Access (DCL commands), Role, Privileges, Creating a database user, GRANT, REVOKE
13	Start of a Lab Project , Introduction to Procedural SQL (PL/SQL), Sections of a PL/SQL block, Variable declaration and initialization, SELECT statement in PL/SQL, Arithmetic expressions
14	PL/SQL: Selection, Repetition, Exception Handling, Cursors
15	PL/SQL: Stored Procedures and Functions, Introduction to Database Triggers
16	Finish of a Lab Project , Database Triggers, Revision